HW 1

1. Inverse Transform Method

- F strictly increasing
- truncated density $f^*(x) = \frac{f(x)}{F(b)-F(a)}$ if $a \leq x \leq b$. 0 ow
- truncated distribution function $F^*(x) = \frac{F(x)-F(a)}{F(b)-F(a)}$ if $a \leq x \leq b$. 1 if x > b, 0 ow

a) show X has distribution function F*

Since F is continuous and strictly increasing, F^{-1} is too.

$X \leq x \iff F^{-1}(V) \leq x \iff V \leq F(x)$

We solve for

$$\begin{aligned} P(X \leq x) &= P(V \leq F(x)) \\ &= P(F(a) + [F(b) - F(a)]U \leq F(x)) \\ &= P(U \leq \frac{F(x) - F(a)}{F(b) - F(a)}) \\ &= \frac{F(x) - F(a)}{F(b) - F(a)} \quad \text{since } P(U \leq u) = u \text{ for } u \in [0,1] \end{aligned}$$

We also check the edges:

- $x < a$ - since $V = F(a) + [F(b) - F(a)]U \geq F(a)$ always ($U \geq 0$), and $F(x) < F(a)$ for x < a (F increasing), we get $V \leq F(x)$ is impossible. So $P(X \leq x) = 0$.
- $x > b$ - $V \leq F(b)$ always since $U \leq 1$. Also $F(x) \geq F(b)$, so $V \leq F(x)$ always holds. We get $P(X \leq x) = 1$.

b) another generation algorithm

Generate $U \sim U(0,1)$
If $F(a) \leq U \leq F(b)$, return $X = F^{-1}(U)$.

$$\begin{aligned} P(X \leq x) &= P(U \leq F(x) \mid F(a) \leq U \leq F(b)) \\ &= \frac{F(x) - F(a)}{F(b) - F(a)} \\ &= F^*(x) \end{aligned}$$

for $a \leq x \leq b$

c) which algorithm is better?

Algorithm 1 is better in terms of computational effort. It requires only one uniform draw and no rejection loop, always producing a usable value. Each trial succeeds with probability $p = F(b) - F(a)$. So on average $1/p$ trials are needed, which is less efficient.

2.

See code

3.

a)

Success probability:

$$Pr(S) = \int_a^b Pr(U \leq \frac{f(y)}{\lambda} \mid Y = y)dy = \int_a^b \frac{f(y)}{\lambda} \cdot \frac{1}{b-a} dy$$

Since $U \sim \text{Unif}(0,1), Pr(U \leq c) = c$ for $c \in [0,1]$.
So

$$Pr(S) = \frac{1}{\lambda(b-a)} \int_b^a f(y)dy = \frac{1}{\lambda(b-a)}$$

b)

$$Pr(Y \leq y \mid S) = \frac{Pr(Y \leq y, S)}{Pr(S)}$$

$$Pr(Y \leq y, S) = \int_a^y \frac{f(t)}{\lambda} \cdot \frac{1}{b-a} dt = \frac{F(y)}{\lambda(b-a)}$$

Using part (a), we divide by Pr(S) and get:

$$Pr(Y \leq y \mid S) = F(y).$$

If we differentiate both sides we get that the conditional density of Y given S is f.

c)

Since e^{1-x} is strictly decreasing, f is decreasing on $[0,1]$, so its maximum is at $x = 0$:

$$\lambda = f(0) = \frac{e-1}{e-2}$$

$$Pr(S) = \frac{e-2}{e-1}$$

4.

a)

$$f(x) = \begin{cases} 3x^2/2, & \text{if } -1 \le x \le 1, \\ 0, & \text{o.w.} \end{cases}$$

Inverse Transform

$$F(x) = \int_{-1}^{1} \frac{3t^2}{2} dt = \frac{x^3+1}{2}, \quad -1 \le x \le 1$$

Set $U = F(X)$ and solve:

$$X = (2U-1)^{1/3}$$

Algorithm:

- generate $U \sim \text{Unif}(0,1)$
- return $X = (2U-1)^{1/3}$

Composition
Using f's symmetry about 0, write $|X|$ as having density $g(x) = 3x^2$ on $[0,1]$ and assign a random sign (± 1) with probability 0.5 each.
Inverting g : $G(x) = x^3 \Rightarrow |X| = U_1^{1/3}$. Then multiply by a random sign with probability 0.5.

Algorithm:

- Generate $U_1, U_2 \sim \text{Unif}(0,1)$, set $R = U_1^{1/3}$
- return $X = R$ if $U_2 < 1/2$, else $X = -R$

Acceptance-rejection
Use $g(x) = \text{Unif}(-1,1)$ as the envelope, $\lambda = max f = 3/2$. From 3a,
$Pr(S) = 1/[\lambda(b-a)] = 1/(3/2 \cdot 2) = 1/3$, so on average 3 trials are needed per accepted value.

The inverse transform method is best here since the cdf inverts with a single cube root, only one uniform draw and no rejection loop.

b)

$$f(x) = \begin{cases} \dfrac{x}{a(1-a)}, & 0 \le x \le a \\ \dfrac{1}{1-a}, & a \le x \le 1-a \qquad 0 < a < \frac{1}{2} \\ \dfrac{1-x}{a(1-a)}, & 1-a \le x \le 1 \end{cases}$$

This is a symmetric trapezoid: a linear ramp up on $[0, a]$, a flat plateau of height $\frac{1}{1-a}$ on $[a, 1-a]$, and a linear ramp down on $[1-a, 1]$.

this is exactly the density of the sum of two independent uniforms,

$$X = U_1 + U_2, \qquad U_1 \sim \mathrm{Unif}(0, a), \quad U_2 \sim \mathrm{Unif}(0, 1-a)$$

(the convolution of two uniforms with supports of length a and $1-a$ produces a trapezoid with ramp width $\min(a, 1-a) = a$ and plateau height $1/\max(a, 1-a) = 1/(1-a)$, matching f exactly).

Composition

- Generate $U_1 \sim \mathrm{Unif}(0, a)$, $U_2 \sim \mathrm{Unif}(0, 1-a)$, independently
- Return $X = U_1 + U_2$
  This is exact, requires only two uniform draws and one addition: no inversion, no rejection loop.

Inverse Transform

The cdf must be built piecewise:

$$F(x) = \begin{cases} \dfrac{x^2}{2a(1-a)}, & 0 \le x \le a \\ \dfrac{a}{2(1-a)} + \dfrac{x-a}{1-a}, & a \le x \le 1-a \\ 1 - \dfrac{(1-x)^2}{2a(1-a)}, & 1-a \le x \le 1 \end{cases}$$

Algorithm:

- Generate $U \sim \mathrm{Unif}(0, 1)$
- Compare U against the breakpoints $F(a) = \dfrac{a}{2(1-a)}$ and $F(1-a) = 1 - \dfrac{a}{2(1-a)}$ to determine which branch applies
- Solve the corresponding equation for X (quadratic on the ramps, linear on the plateau)
  This works but requires piecewise case logic and solving a quadratic on two of the three branches.

Acceptance–Rejection

Use envelope $g(x) = \mathrm{Unif}(0, 1)$, $\lambda = \max f = \dfrac{1}{1-a}$. Acceptance probability:

$$\Pr(S) = \frac{1}{\lambda} = 1 - a \in \left(\tfrac{1}{2}, 1\right)$$

Reasonably efficient, but still needs a rejection loop, averaging $\frac{1}{1-a} > 1$ trials per accepted value.

The composition method is best: it is exact, needs only two uniform draws and a single addition, with no branching or rejection. Inverse transform is workable but requires piecewise logic and solving quadratics; acceptance–rejection is simple to code but wastes draws for no real benefit over composition.